

Name:	Tutor group:
Teacher:	Science set:

**Pearson Edexcel IGCSE
Double Award Science
Island School Chemistry UNIT 1**

Matter and Separation Techniques

(Y10 2026-2027)

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Lesson 1 - States of Matter

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Revision worksheets

UNIT 1 Syllabus Statements

Lesson	Syllabus statement from the exam board	Textbook pages
1 States of Matter	1.1 understand the three states of matter in terms of the arrangement, movement and energy of the particles. 1.2 understand the interconversions between the three states of matter in terms of: • the names of the interconversions • how they are achieved • the changes in arrangement, movement and energy of the particles.	Read pages 6 - 8 Study Questions on page 8
2 Diffusion and Dissolving	1.3 understand how the results of experiments involving the dilution of coloured solutions and diffusion of gases can be explained. 1.4 know what is meant by the terms: • solvent • solute • solution • saturated solution.	Read pages 2 - 5 Study Questions on page 5
3 Elements, Compounds and Mixtures	1.8 understand how to classify a substance as an element, compound or mixture 1.9 understand that a pure substance has a fixed melting and boiling point, but that a mixture may melt or boil over a range of temperatures. 1.14 know what is meant by the terms atom and molecule	Read pages 9 - 11 Study Questions on page 11
4 Filtration and Crystallisation	1.14 know what is meant by the terms atom and molecule 1.10 describe these experimental techniques for the separation of mixtures: • filtration • crystallisation	Read pages 12 - 17 Study Questions on page 17
5 Distillation	1.10 describe these experimental techniques for the separation of mixtures: • simple distillation • fractional distillation	
6 Chromatography	1.10 describe these experimental techniques for the separation of mixtures: • paper chromatography. 1.11 understand how a chromatogram provides information about the composition of a mixture 1.12 understand how to use the calculation of R_f values to identify the components of a mixture 1.13 practical: investigate paper chromatography using inks/food colourings	

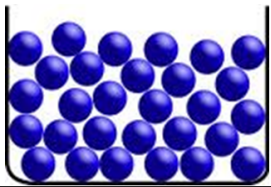
Unit 1 Glossary

By the end of this unit, you should be familiar with the following terms. You will need to correctly use these words during any assessments.

Atom	The smallest part of an element that can exist.
Particles	Extremely small pieces of matter, for example, atoms, molecules and ions.
Molecule	A molecule consists of an exact number of atoms joined together covalently.
Element	A pure substance that cannot be split up into anything simpler. An element contains just one type of atom.
Symbols	Chemists use symbols to represent atoms. Every element has a different symbol. For example, the symbol for carbon is C.
Diffusion	The random movement and mixing of particles in a liquid or gas.
Dissolving	Solid particles break apart and exist in the gaps between the liquid particles.
Simple distillation	A separation technique that allows a pure liquid to be separated from a mixture. Involves the liquid boiling and then condensing.
Filtration	A separation technique that allows an insoluble solid to be separated from a mixture in a liquid. The liquid passes through the filter paper but the solid does not.
Crystallisation	A separation technique that allows a soluble solid to be separated from a dissolved mixture in a liquid. The mixture is heated and the liquid partially evaporates. As the mixture cools, the solid starts to appear. This is called crystallisation.
Paper chromatography	A technique that allows a complex mixture of substances to be separated and identified. Each substance rises up the filter paper by different distances.
Retention factor (R_f value)	Used to identify the substances used in paper chromatography. The distance moved by the solute / distance moved by the solvent.
Solute	Solid particles that you are trying to dissolve.
Solvent	Liquid that you are using to dissolve the solid. Water is the most common solvent, but there are others such as propanone is used to dissolve nail polish.
Solution	When the solid has dissolved in the liquid, we call this a solution.
Saturated solution	This is the maximum amount of solute that a solvent can dissolve at a given temperature.
Solubility	This is a measure of how much solute has been dissolved in the solvent. It is often measured in g of solute per 100g of solvent at a specific temperature)

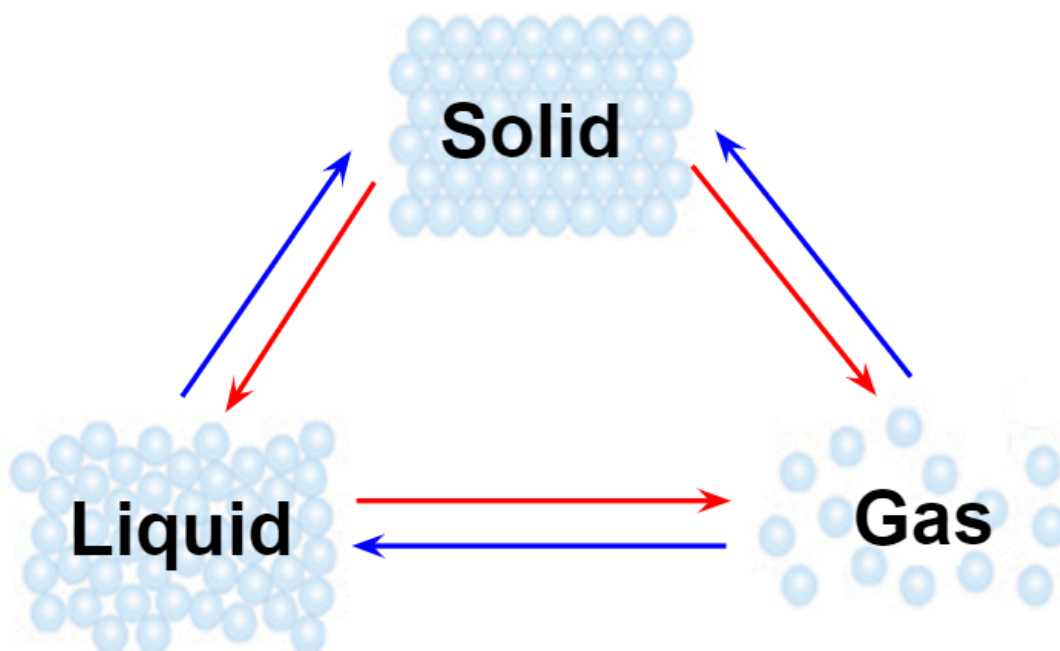
States of Matter

1. Complete the following table to summarise the properties of solids, liquids and gases (some have been done for you)

State	Solid	Liquid	Gas
Arrangement	Very close together Regular arrangement		
Motion		Slip and slide over each other	
Properties (shape, volume & compressibility)	Definite shape Not compressible	Definite volume	Spread out to fill available volume Volume depends on the container Compressible
Model of particles			

2. a) The particles in which of the 3 states of matter have the most energy?
- b) In which states of matter are particles touching? and
- c) The particles in which of the 3 states of matter are furthest apart?

3. Solids, liquids and gases can be interconverted by either heating or cooling. Complete the following diagram by adding the names for the interconversions on the appropriate arrows.



4. What is the difference between evaporation and boiling?
-
-

5. Convert the following statements to chemical equations:

a) Water turning into steam

.....

b) Iron (Fe) melting

.....

c) Dry ice turning into carbon dioxide gas

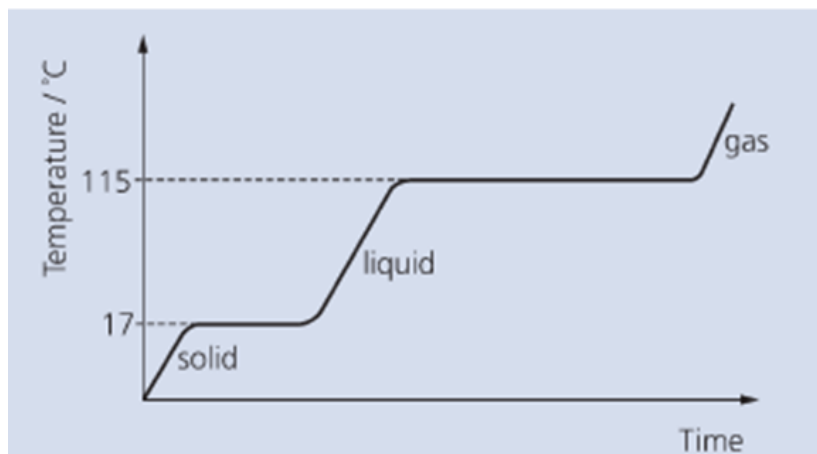
.....

d) Liquifying oxygen from the air

.....

States of Matter

1. Below is a heating curve for a pure substance. It shows how the temperature rises over time, when the substance is heated until it melts, then boils.



- a) What is the melting point of the substance?
- b) What happens to the temperature while the substance changes state?
.....
.....
- c) The graph shows that the substance takes longer to boil than to melt. Suggest a reason for this.
.....
.....
.....
- d) How can you tell that the substance is not water?
- e) What is the state of this substance at room temperature (25°C)?
- f) Carefully sketch a cooling curve for this substance

g) Describe the *change* in **arrangement**, **movement** and **energy** of the particles as they change state from solid to liquid.

Arrangement:.....

Movement:

Energy:

h) Using information from the first table below, complete the second table with the words 'solid', 'liquid' or 'gas'

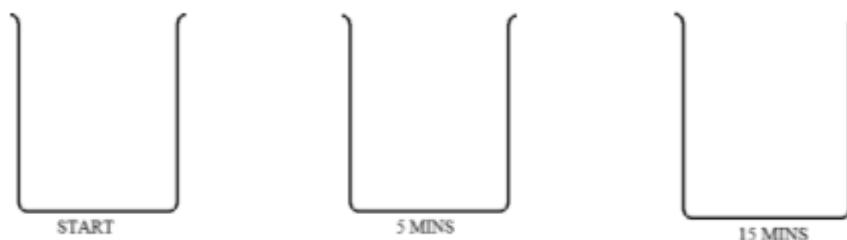
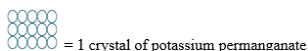
Substance	Melting Point (°C)	Boiling point (°C)
water	0	100
methane	-182	-162
oxygen	-219	-183
ethanol	-117	79

Substance	State at 25°C	State at 1000°C	State at -50°C
water			
methane			
oxygen			
ethanol			

Dissolving & Diffusion

Method: Carefully put a purple crystal of potassium permanganate (**solute**) into a beaker of water (**solvent**) using a hollow glass tube. Do not move the beaker but check it after 5 and 15 minutes

Draw three particle diagram below to explain what is happening to the **solution over time**.

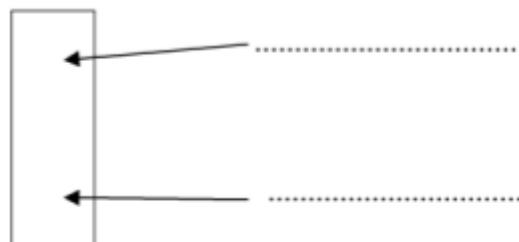
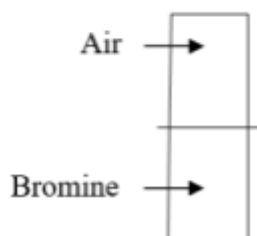


2. Explain what is happening to the particles in your diagram. Try to use as many of the following words as possible.

solvent	diffusion	move	particles	solution	dissolves	spreads	water	solute
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3. How could you speed up the processes taking place in the beaker in the experiment above? Explain why this would speed it up in terms of the **energy** of the particles involved.

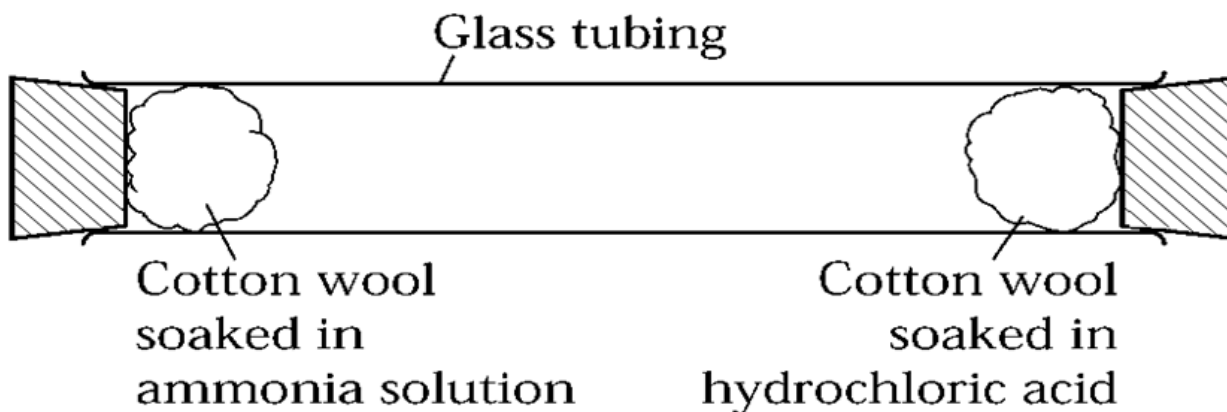
4 Label the following gas jars after the experiment after the cover slip is removed:



5 Diffusion happens faster in gases than in liquids. Explain why this happens.

Diffusion of Gases

In this experiment (demo) you will investigate what happens when two different gases diffuse through the air. Hydrogen chloride and ammonia will react together to give a white solid. When the two gases meet, this appears as white smoke in the air. Particles of hydrogen chloride are about twice as heavy as particles of ammonia.



Recording your results

1. When your teacher puts the cotton wool in the tube, look carefully to see where the white smoke ring appears. *Mark this clearly on the diagram.*

Considering your results/conclusions

2. Which gas moved faster, ammonia or hydrogen chloride?
3. Explain how you can tell this from the results
4. What would happen to the time taken to form the white smoke ring if the tube was heated? And where would the ring form? Explain why.
.....
.....
.....
5. *On the above diagram* draw a line to illustrate the path taken by an ammonia molecule and hydrogen chloride molecule as they move along the tube
6. The particles in a gas move at a very fast speed. Explain why it took some time for the white smoke ring to form.
.....
.....

Extension: Give an equation to show the reaction between HCl and NH_3 to give a white smoke. Given that $\text{KE} = \frac{1}{2}mv^2$, explain why gases with a smaller mass tend to diffuse faster at the same temperature. Use the internet to research and explain this.

Elements, Compounds & Mixtures

Element names and symbols:

- | | |
|----|-----|
| 1) | 6) |
| 2) | 7) |
| 3) | 8) |
| 4) | 9) |
| 5) | 10) |

Define the term **element**:

Particle diagrams for a solid, liquid and gaseous element:

Define the term **compound**:

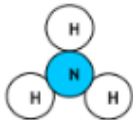
What does the formula H_2O mean?

Define the term **molecule**:

Particle diagrams for H₂O in the solid, liquid and gaseous states:

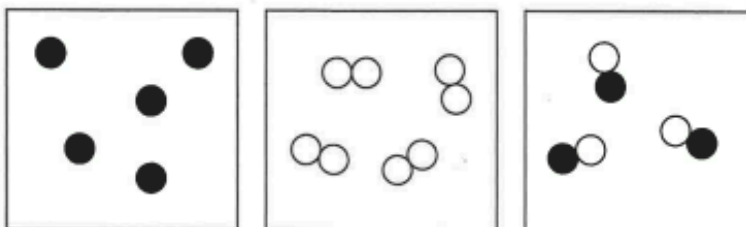


Fill in the blanks

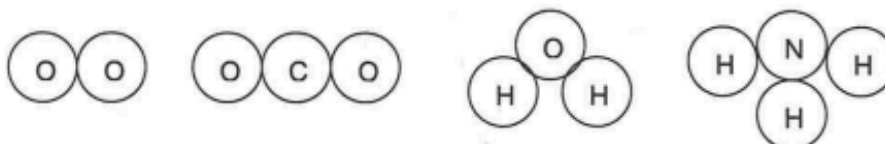
Name of substance	Diagram of molecule	Chemical formula	Element or compound?
Water			
Nitrogen			
Carbon dioxide			
Ammonia			
Sulfur dioxide			
Hydrogen			
Hydrogen chloride			
Oxygen			

Questions about elements and compounds

- Which of the boxes contains:
 - Molecules of an element?
 - Molecules of a compound?
 - Atoms of an element?
 - Do any of the boxes contain a mixture? Explain your answer.



- Look at the following substances. Circle the elements and underline the compounds
 - Li
 - N₂
 - CO
 - Cl₂
 - S₈
 - NaCl
- How many elements make up each compound?
 - HCl
 - CaCO₃
 - HNO₃
 - H₂SO₄
 - CH₃OH
- How many atoms are there in each of these molecules?
 - N₂
 - CO₂
 - O₃
 - CH₄
 - SO₂
 - C₂H₄
- Circle the molecule with the most atoms, in each pair;
 - HCl or POCl
 - NH₃ or HNO₃
 - C₃H₈ or C₂H₅OH
 - CH₃COCH₃ or C₆H₆
- What is the formula for each molecules shown below



Define the term **mixture**?

Draw three particle diagrams to show the following:

A solid mixture (e.g. a metal alloy)

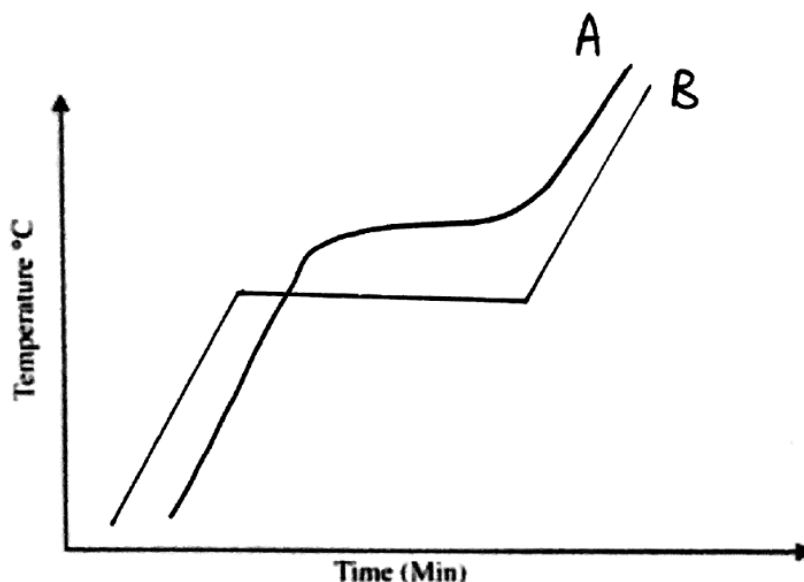
A 'liquid' mixture (e.g. seawater)

A gas mixture (e.g. air)

Pure vs. Impure

1. In your own words, describe the difference between **pure** and **impure**.

2. The curves below were obtained by a student after heating two solid substances A & B.



a) Which substance (A or B) has the higher melting point?

b) Explain which curve represents an impure substance?
.....
.....

3. According to our definitions H_2O (water) is a compound. However, it is sometimes called a molecule. O_2 (oxygen gas) is also a molecule. Use this sentence to define a molecule.

A _____ consists of _____ or more _____ chemically _____ together. These _____ can be of the same _____ or different elements.

Extension: There are around 100 different elements. How many compounds do you think there are? (Hint: think about how many different combinations you could make with just 3 different elements.)

Blank page for notes about atoms, elements, compounds and mixtures

**Blank page for notes about atoms, elements,
compounds and mixtures**

Filtration & Crystallisation

Aim: To design and carry out practical work to try and separate pure samples of substances from a range of mixtures.

Safety: Goggles must be worn. Be careful using hot glassware. 

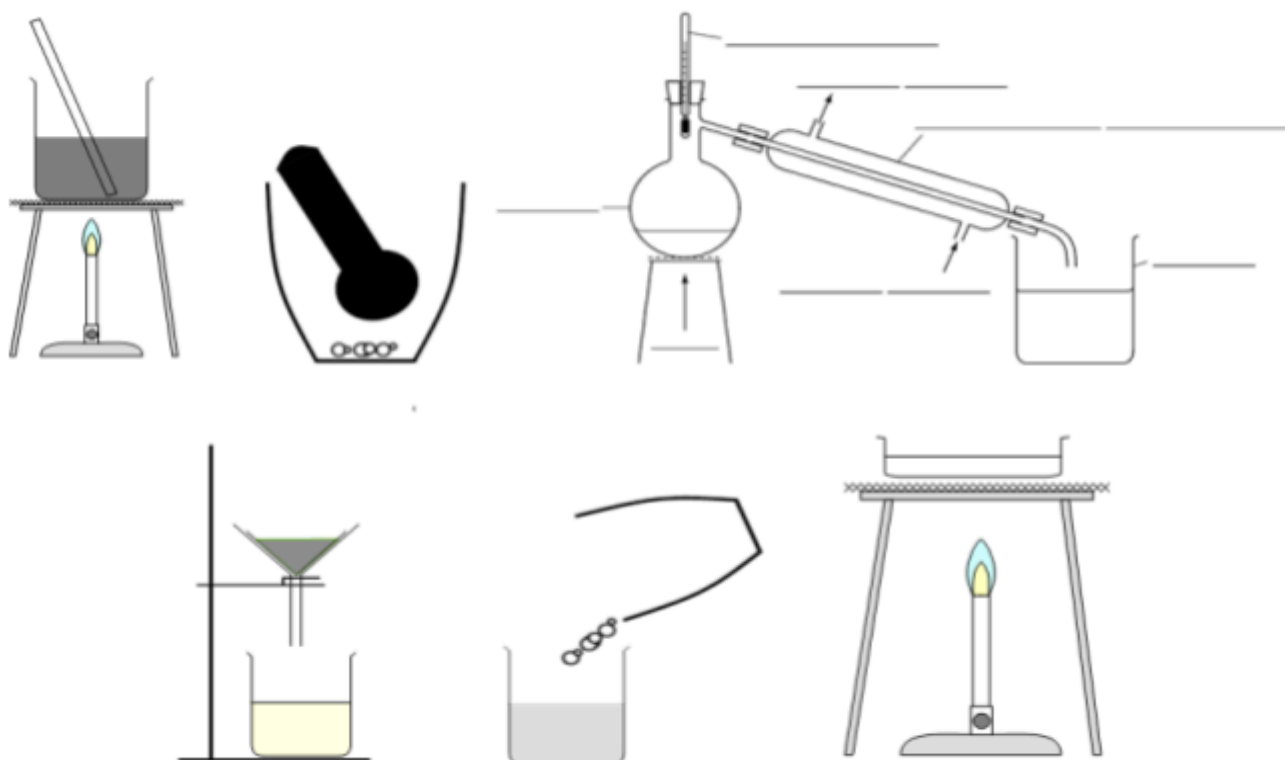
Design/Plan:

You have a practical problem to solve in your team of 2 or 3 students. You need to design an experiment to separate a pure sample of salt from the rock salt.

- Brainstorm your plan on the additional note space on the next page.
- Draw labelled diagrams showing how you will need to set up the apparatus and explain what happens at each stage.
- Write a method using numbered points to explain what you will do.

Some of the apparatus below may help you decide what to do.

Diagrams and Apparatus:



Planning Notes - design an experiment to separate a pure sample of salt from the rock salt.

Analysis / Conclusion

1) Explain (in terms of the particles involved) how you were able to separate a pure sample of salt from the rock salt.

2) If you started with 5g of rock salt and weighed the mass of pure salt obtained at the end and it was 4g, what % of salt is there in rock salt?

3) A student carried out this practical and the rock salt obtained was grey rather than pure white. What does this suggest and what could the student do next to test if the salt is pure?

4) What is the name of the process called when the crystals of pure salt start to appear? If a student weighed their salt crystals before the water had all evaporated away, would this be an accurate mass of salt? What should they do to improve the accuracy of the mass?

5) Henrique spilled some water onto a black kitchen worktop and did not wipe up the spill. The water looked clear. Eventually, the water dried but a white mark was left. What does this tell you about the water?

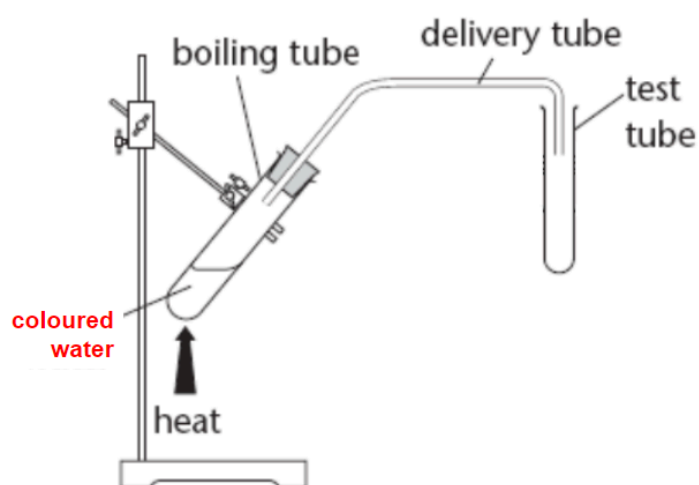
	True	False	Insufficient evidence
The water was mixed with a white substance			
The water was pure			
The water contained a dissolved substance			
The water was salty			

Distillation

Aim: Carry out practical work to separate a pure sample of water from some inky water.

Safety: Goggles must be worn. Be careful using hot glassware. 

What is distillation?



What is the main issue with this apparatus?

Draw a labelled diagram of some better apparatus including a Liebig condenser.

Analysis/conclusion:

- 1) Explain (in terms of the particles involved) how you were able to separate a pure sample water from your mixture.
- 2) How could a student test the purity of the water they obtained from this experiment?
- 3) You can buy 'distilled' water from many shops. What does this tell us about the purity of the water that we are buying?
- 4) Why do we buy 'mineral' water instead of 'distilled' water? What are the advantages of each?
- 5) How is fractional distillation is different from simple distillation.
- 6) A mixture of three liquids needs separating. The expected temperature range for collecting each liquid in the distillate is shown below.

liquid X: 40°C-100°C

liquid Y: 120°C-160°C

liquid Z: 180°C -200°C

The mixture was added to the distillation apparatus and was heated quickly to 150°C. Select the row of the table that shows what would be collected in the distillate.

	liquid X	liquid Y	liquid Z
A	✓	✓	✓
B	✓	X	X
C	✓	✓	X
D	X	✓	X

Chromatography

Aim: Describe these experimental techniques for the separation of mixtures using paper chromatography. Understand how a chromatogram provides information about the composition of a mixture and how to use the calculation of R_f values to identify the components of a mixture.

Apparatus: Strips of chromatography paper, beakers, felt tip pens (dark colours, brown, black, purple green). Plus an unknown black pen of a different brand.

Method:

1. Draw a pencil line about 1.5 cm from the bottom of the strip of chromatography paper.
2. Put a small mark of the felt tip on the line on this line of the chromatography paper.
3. Repeat with other colours of felt tip about 2 - 3 cm from the previous spot.
4. Label each spot in pencil underneath the line.
5. Hang the chromatography paper using a splint in a beaker. (If needed, you can use a paperclip to hold it in place.) Put the paper in a beaker containing 1 cm depth of water, making sure the pencil line is about 0.5 cm above the level of the water and the paper doesn't touch the sides of the beaker.
6. Allow the water (solvent) to rise up the paper. The greater the time for separation the more accurate the results. This should take about 10-15 mins but you need to keep checking it.
7. Remove the paper before the solvent reaches the top, mark the solvent (water) line (solvent front) with a pencil.
8. Allow the paper to dry and record the distances that the different coloured dyes have moved with a ruler measuring in mm and put this into the table.
9. Determine the retention factors for each of the coloured dyes obtained using the following equation:

$$\text{Retention factor } (R_f) = \frac{\text{distance moved by solute (dye)}}{\text{distance moved by the solvent (water)}}$$
10. On a separate piece of chromatography paper, set up another chromatogram for the unknown ink in the pen given. Determine the retention factor of any dyes in this ink sample.

Results: Distance moved by the solvent (water) = _____ mm

Colour	Distance (mm)	Retention factor (R_f)

Unknown ink:

Distance moved by the solvent (water) = _____ mm

Determine the retention factors: R_f s = _____

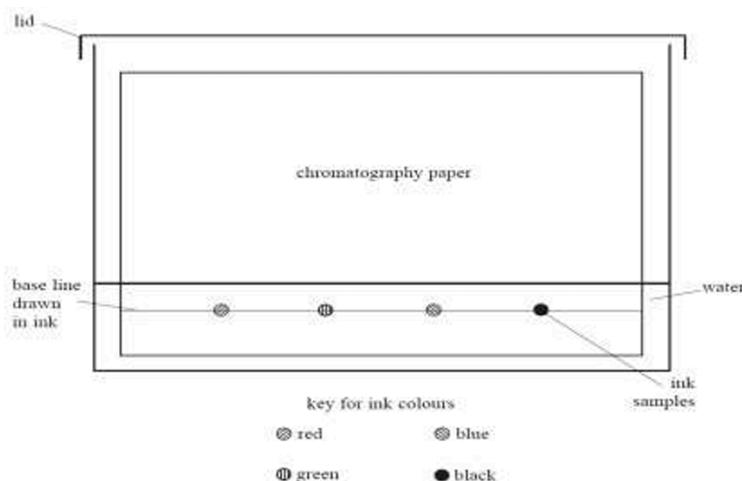
Analysis/Conclusion:

1. Which colour of felt tip contained the most different dyes?
2. Do any colours contain only one dye? If the ink was made of only one pure dye, how many spots would you expect to see?
3. Did any of the ink from the felt tips not move up the filter paper? How could this happen?
4. What would happen if the level of the water (solvent) was higher than the ink at the start of the experiment?
5. Did the unknown felt tip pen contain any of the same dyes in the ink as the other samples? How can you tell?

Extension: How could you improve the accuracy of this experiment? Explain why some dyes move greater distances on the paper than others.

Chromatography Exam Questions

1. A student investigates the colours contained in inks from felt-tip pens. He uses chromatography and sets up his experiment as shown:



(a) Identify two mistakes in the way he sets up the experiment. For each mistake, state what problem it would cause.

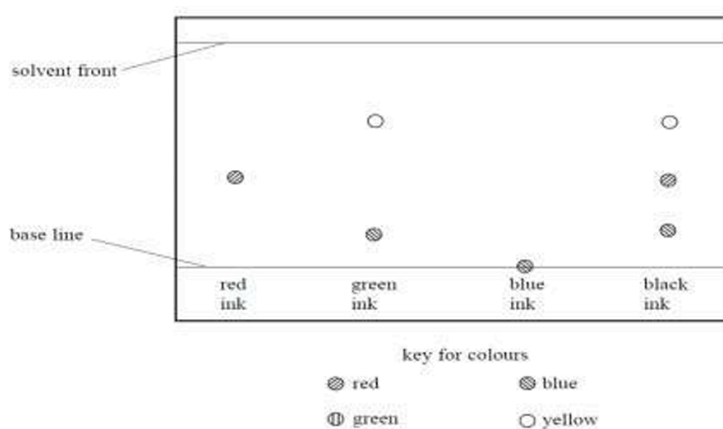
1st mistake = _____

Problem = _____

2nd mistake = _____

Problem = _____

(b) Another student repeats the experiment, but does not make any mistakes. She uses inks from four different felt tip pens. The diagram shows her results.

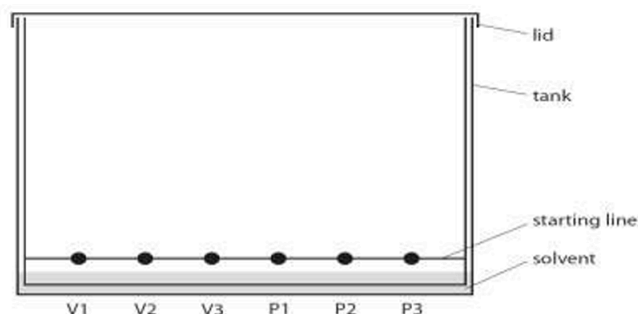


(c) (i) How many different colours does the black ink contain? _____

(ii) Which of the inks tested could be mixed together to make the black ink? _____

(iii) Which of the inks tested is insoluble in water? Explain your answer. _____

2. A student investigates the pigments found in some vegetables and fruit. She obtains some coloured vegetable and fruit extracts from carrots, tomatoes and sweet potatoes. She places a spot of each extract on chromatography paper, along with spots of the three pigments beta-carotene, chlorophyll and lycopene. Her teacher provides a solvent containing volatile, flammable, organic compounds for the experiment. The diagram shows the apparatus at the start of the experiment:



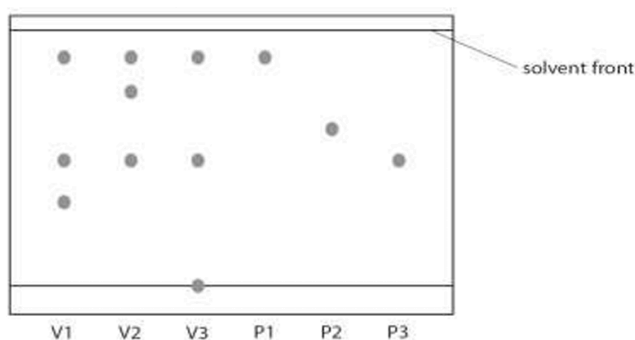
Key to vegetable and fruit extracts and pigments

V1 = carrots V2 = tomatoes V3 = sweet potatoes
P1 = beta-carotene P2 = chlorophyll P3 = lycopene

(a) Explain why it is important for the solvent level to be below the spots. _____

(b) State two potential problems that are prevented by fitting the tank with a lid. _____

The diagram shows the chromatogram at the end of the experiment.



(c) Put a tick next to **three** statements that are supported by the chromatogram.

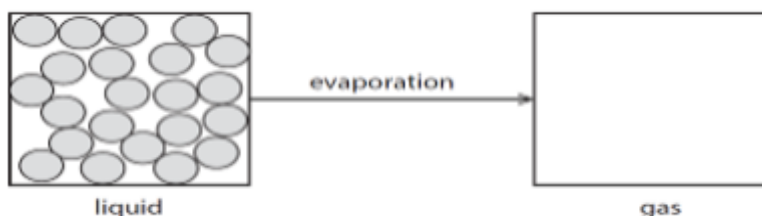
- | | |
|--------------------------|---|
| ☐ | Chlorophyll is not present in carrots, sweet potatoes or tomatoes |
| ☐ | Beta-carotene is present in carrots but not present in tomatoes. |
| ☐ | Both beta-carotene and lycopene are present in sweet potatoes. |
| ☐ | Lycopene is present in tomatoes but not present in carrots. |
| ☐ | Both carrots and tomatoes contain a pigment other than beta-carotene, chlorophyll and lycopene. |

(d) Suggest a reason why there is a spot on the starting line in the chromatogram for sweet potatoes. _____

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Questions on Matter and Separating Techniques

1. When a liquid evaporates at room temperature, it changes into a gas.
The diagram shows the arrangement of the particles in a liquid.



(a) Complete the diagram to show the arrangement of four particles in a gas. (1)

(b) Describe the movement of particles in a gas. (1)

(c) Explain why heating a liquid causes it to evaporate more quickly. (2)

2. Techniques used in the separation of mixtures include

A = crystallisation

B = filtration

C = fractional distillation

D = simple distillation

For each separation, select the most suitable technique, A, B, C or D, used to obtain the first named substance from the mixture. Each letter may be used once, more than once or not at all.

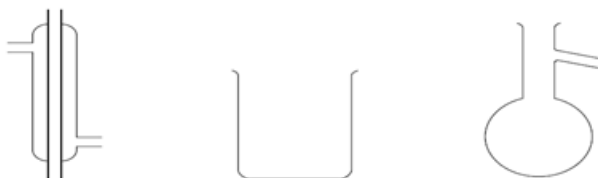
(a) Pure water from sea water =

(b) Ethanol from a mixture of ethanol and water =

(c) Calcium carbonate from a mixture of calcium carbonate and water =

(d) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ from $\text{CuSO}_4(\text{aq})$ = (4)

(e) Sea water contains dissolved sodium chloride. The following pieces of laboratory apparatus can be used to make drinking water from sea water.



(i) Draw a labelled diagram to show how these pieces of apparatus can be assembled to do this. (3)

(ii) Name the technique used in this process. (1)

3. A teacher uses this apparatus to show the reaction between ammonia gas and hydrogen chloride gas to form ammonium chloride.



Concentrated ammonia solution gives off ammonia gas. Concentrated hydrochloric acid gives off hydrogen chloride gas. The reaction between the gases can be represented by this equation.



The diagram below shows the apparatus after a few minutes.



- (a) Which process occurs when the gas particles move along the tube? (1)
- (b) Name the white solid that forms in the tube. (1)
- (c) Explain how the diagram shows which gas moves more quickly during the experiment. (1)
- (d) The particles in a gas move very quickly. Why does it take a few minutes for the white gas to form? (1)

4. This is a description of how the orange colouring can be extracted from rose petals.

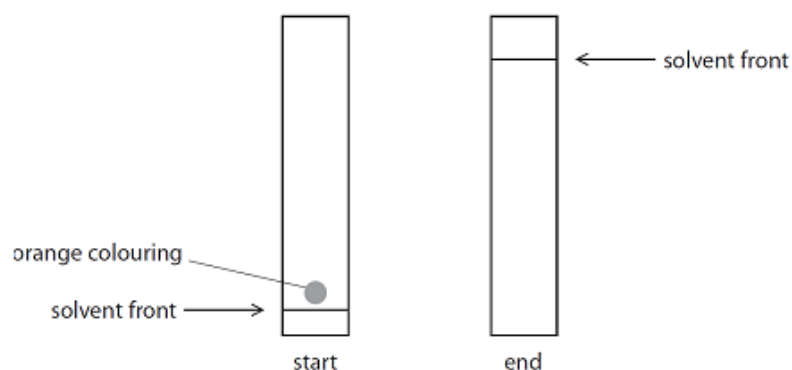
- crush the petals using a pestle and mortar
- add the crushed petals to some ethanol in a beaker
- heat to about 60°C and stir to produce an orange solution
- separate the orange solution from the petals

(a) (i) Suggest why ethanol is used instead of water. (1)

(ii) Ethanol is a flammable liquid. Suggest how it could be heated safely. (1)

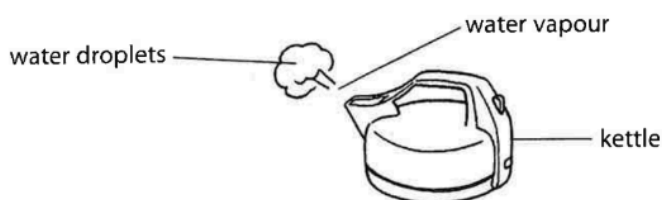
(iii) How could the orange solution be separated from the petals? (1)

(b) The orange colouring is analysed using chromatography and is found to consist of two different colours, red and yellow. The diagram shows the chromatography paper at the start of the experiment. Complete the diagram to show a possible result at the end of the experiment. (2)



5.

The diagram shows a kettle of boiling water.



As the water vapour cools it turns into droplets of liquid water.

(a) The change of state when water vapour changes into liquid water is described as (1)

- ☐ A boiling
- ☐ B condensation
- ☐ C evaporation
- ☐ D sublimation

(b) Describe what happens when water vapour cools to form liquid water.

Your answer should include the change in the energy, arrangement and movement of the particles.

(3)

change in energy

.....

.....

change in arrangement

.....

.....

change in movement

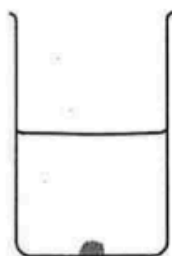
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6.

Hydrated copper(II) sulfate is a soluble blue solid. A large crystal of this solid is placed at the bottom of a beaker of water.

The diagram shows the beaker immediately after placing the crystal in it, and after two days.



after placing the crystal



after two days

- (a) After two days, the crystal becomes smaller and the liquid near the bottom of the beaker becomes blue.

Which statement explains these observations?

(1)

- | | |
|--|---|
| ☐ A the crystal dissolves | ☐ C the crystal melts |
| ☐ B the crystal freezes | ☐ D the crystal sublimates |

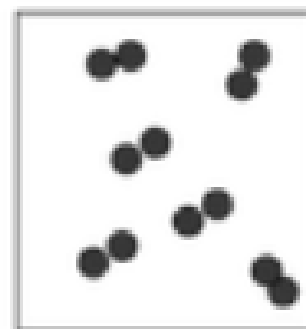
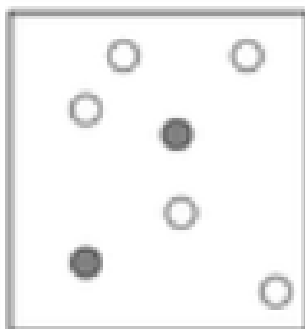
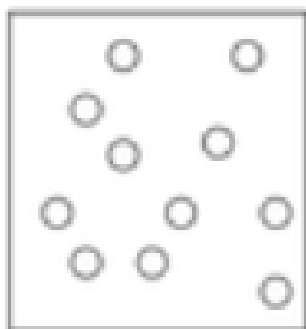
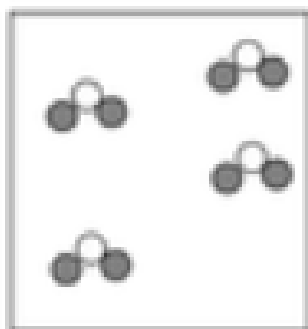
- (b) After two weeks, the crystal has disappeared.

Which statement best describes the appearance of the liquid in the beaker after two weeks?

(1)

- | | |
|--|--|
| ☐ A it is all blue | ☐ C only the lower part is blue |
| ☐ B it is all brown | ☐ D only the upper part is blue |

- (c) Describe each of the following boxes as either 'element', 'compound' or 'mixture'.

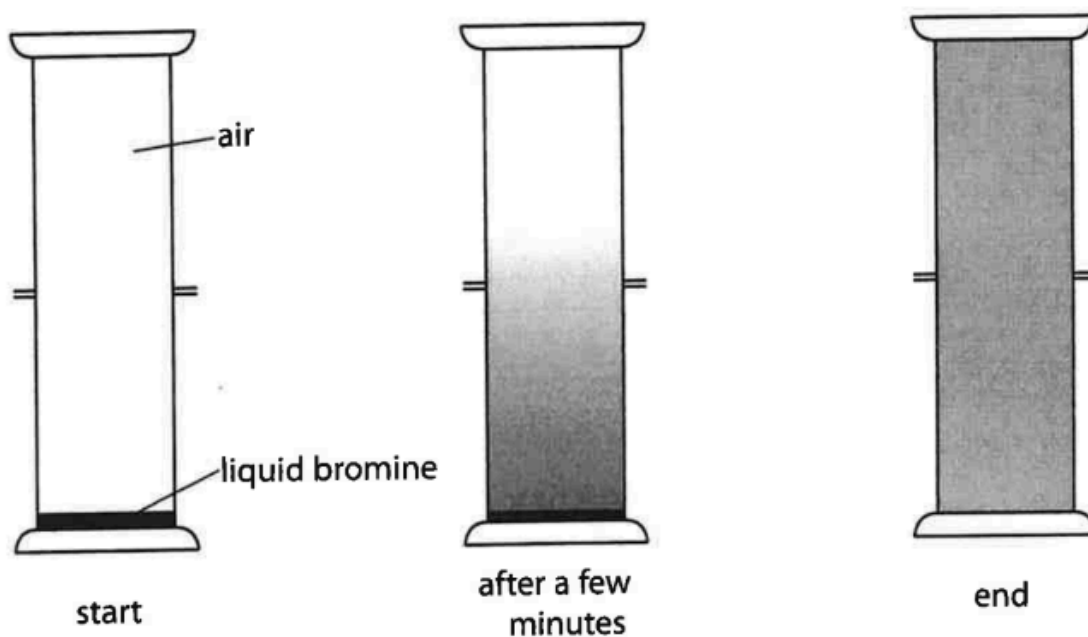


7.

A teacher demonstrates, in a fume cupboard, two experiments to show the movement of particles.

- (a) In the first experiment she places some liquid bromine at the bottom of a gas jar. She then places another gas jar containing air on top of it, as shown in the diagram.

The diagram shows the apparatus at the start, after a few minutes and at the end of the experiment.



Place crosses (☒) in **two** boxes to show which statements are correct about this experiment.

- ☐ **A** All the air particles in the upper gas jar stay there.
- ☐ **B** Bromine and air react to form bromine oxide.
- ☐ **C** Bromine has a darker colour than air.
- ☐ **D** Bromine vapour diffuses upwards.
- ☐ **E** Liquid bromine sublimates during the experiment.
- ☐ **F** The concentration of bromine in the lower gas jar does not change.

8.

(a) The list shows some techniques used to separate mixtures.

- A** crystallisation
B filtration
C fractional distillation
D paper chromatography
E simple distillation

Complete the table to show the best method of obtaining each substance from the mixture.

In each case, choose one of the letters A, B, C, D or E. Each letter may be used once, more than once or not at all.

(2)

Substance	Mixture	Letter
sand	sand and water	
solid copper(II) sulfate	aqueous copper(II) sulfate	

(b) Gold occurs in ores, which are mixtures of gold and other substances. Several elements and compounds are used in the extraction of gold from its ores.

Each box below represents the substances present in one part of the extraction process.

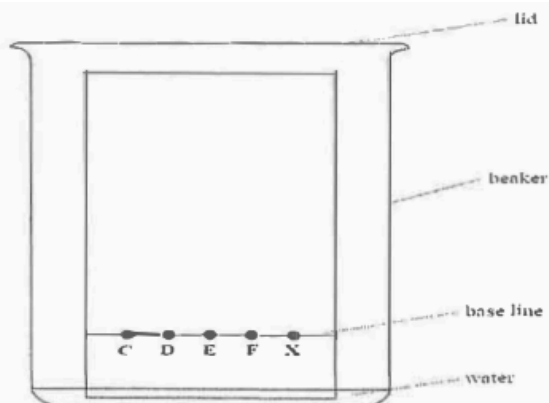
Classify the contents of each box as a compound, an element or a mixture by writing your choice below each box.

(1)

	Au NaCN O₂ H₂O	NaCN NaCN NaCN NaCN	Au H₂O H₂O Au
Compound, element or mixture			

9.

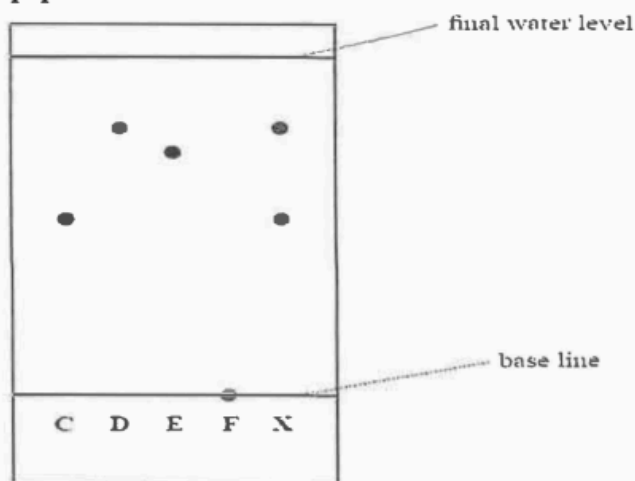
Four Separate food dyes (**C**, **D**, **E** and **F**) and a mixture of food dyes (**X**) were investigated using paper chromatography. The diagram shows the apparatus used:



(a) Why should the water level be below the food dyes? (1)

(b) During the experiment the water rises up the paper. The experiment is stopped just before the water reaches the top of the paper.

The diagram shows the paper after it has been removed from the beaker and dried.



(i) Which of the food dyes **C**, **D**, **E** and **F** does **X** contain? (1)

(ii) Suggest why food dye **F** did not move up the paper during the experiment. (1)

(c) Each food dye has an R_f value.

Record the distances for food dye **D** in the table below and calculate its R_f value.

(3)

Distance moved by food dye D from base line in mm	
Distance moved by solvent front from base line in mm	
R_f value	

Blank page for notes about anything from Unit 1
